

Endurance Scenario – Analysis Report

Overview

The implemented endurance controller integrates:

- Lane Keeping Assist (LKA),
- Autonomous Speed Control,
- and Automatic Emergency Braking (AEB)

into a single autonomous driving system in IPG CarMaker.

The controller performs simultaneous lateral and longitudinal control during:

- straight-road driving,
 - curved-road navigation,
 - and dynamic obstacle handling.
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LKA Analysis

The LKA module uses:

- RSDS camera input,
- OpenCV-based lane detection,
- PID steering control.

The system successfully maintained lane tracking on:

- straight roads,
- and curved-road sections.

Observations

- Stable lane following was achieved throughout the endurance run.
- Steering remained smooth without severe oscillation.
- Vehicle remained inside road boundaries during curve navigation.

Limitation Observed

The current implementation uses:

single-lane edge tracking

which caused:

- persistent steering bias,
 - near-constant steering values on long curves,
 - and continuous camera yaw during sustained turning.
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Autonomous Speed Control Analysis

The longitudinal controller uses:

- PI-based speed regulation,
- throttle control through DVA commands.

Observations

- Vehicle autonomously accelerated to approximately:

30 km/h
- Cruise speed remained stable during normal driving.
- Small speed overshoot occurred after AEB release due to integral windup effects.

Improvement Applied

Integral reset logic was added during braking to reduce post-braking speed overshoot.

AEB Analysis

The AEB system uses:

- radar distance (**Dist**),
- relative velocity (**Vrel**),
- lateral target offset (**DistY**),
- TTC-based collision estimation.

Observations

- Target vehicle was successfully detected during dynamic insertion.
- AEB triggered automatically before collision.
- Throttle was reduced to zero during braking.
- Ego vehicle stopped safely before impact.

Persistent Stop Logic

A stop-hold state machine was implemented to:

- avoid repeated stop-go braking,
- keep the vehicle stopped until the obstacle fully exited the ego lane.

This improved braking stability significantly.

Integrated LKA + AEB Behaviour

During the endurance scenario:

- LKA remained continuously active,
- while AEB temporarily overrode longitudinal control during hazard detection.

Observations

- Vehicle maintained lane tracking even during braking.
 - Smooth transition occurred between:
 - cruise,
 - braking,
 - stop,
 - and resume states.
 - System successfully handled:
 - curved-road driving,
 - obstacle insertion,
 - autonomous stop-and-resume operation.
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Key Limitations Identified

Limitation	Observation
Single-lane tracking	Steering bias on curves
Camera perspective sensitivity	Rotating camera feed during sustained yaw
Simplified radar logic	No object classification or trajectory prediction
Camera perspective sensitivity	Sensitive to lateral offset tuning

Proposed Improvements

- Two-lane lane-center tracking
- Camera-radar sensor fusion
- Adaptive braking control
- Advanced path-tracking algorithms
- Multi-object obstacle handling

Conclusion

The endurance controller successfully demonstrated integrated autonomous driving functionality using:

- camera-based LKA,
- radar-based AEB,
- and autonomous speed regulation.

The vehicle completed:

- straight-road lane keeping,
- curved-road navigation,
- dynamic obstacle detection,
- autonomous braking,
- and safe resume behaviour

within a single continuous endurance scenario.

